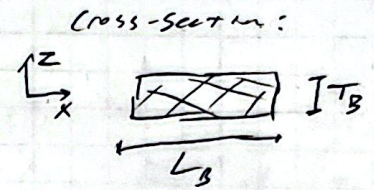
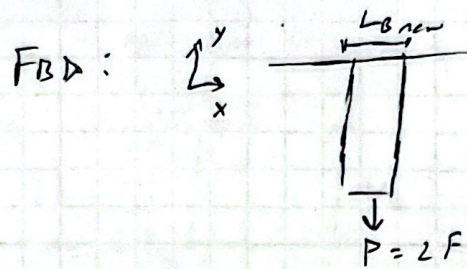


## New Feature B: (STRESS)



## • New assumptions:

- Feature B acts as Axial loaded bar
- Force P pulls uniformly at cross-section
- Assume  $L_{Bnew}$  (in mm) is same as diameter at Feature A  
For Symmetry Assumption  
ON ALL FEATURES POSSIBLE  
( $L_{Bnew} = d_A = 0.798 \text{ in}$ )

- Assumption of uniform Thickness will not take into account on Feature B  
due to reference in Appendix C

Solve for  $A_{Bnew}$  Through cross-section:

$$A_{Bnew} = L_{Bnew} \cdot T_B$$

Solving for  $\sigma_{allow}$ :

$$\sigma_{allow} = \frac{\sigma_{yield}}{FOS}$$

$$\sigma_{allow} = \frac{F}{A} = \frac{P}{A_B} = \frac{P}{L_{Bnew} \cdot T_B} = \frac{2F}{L_{Bnew} \cdot T_B}$$

$$\frac{\sigma_{yield}}{FOS} = \frac{2F}{L_{Bnew} \cdot T_B}$$

Solve for  $T_B$ :

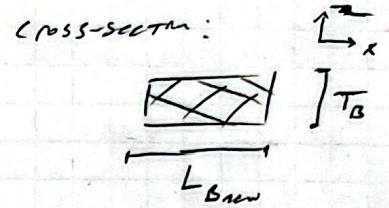
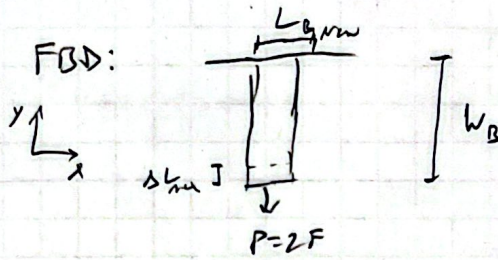
$$T_B = \frac{2F(FOS)}{\sigma_{yield} \cdot L_{Bnew}}$$

$$T_B = \frac{2(500 \text{ lbf})(4)}{(40000 \text{ psi})(0.798 \text{ in})}$$

$$T_B = 0.1253 \text{ in}$$

New Feature B (STRA):

No New Assumptions:



Solve for  $A_B$  from cross-section:

$$A_B = L_{Bnew} \cdot T_B$$

Bending Formula:  $\Delta L = \frac{PL}{AE}$

$$\Delta L_{max} = \frac{2F(W_B)}{A_B(E)} = \frac{2F(W_B)}{T_B(L_{Bnew})E}$$

$$T_{Bstart} = \frac{2F(W_B)}{L_{Bnew} \cdot \Delta L_{max} \cdot E}$$

$$T_{Bstart} = \frac{2(50016N)(0.75in)}{(0.798in)(0.005in)(10 \cdot 10^6psi)}$$

$$T_{Bstart} = 0.0188in$$

Governing  $T_B$ :

$$T_{Bstress} > T_{Bstart}$$

$$(0.1253in) > (0.0188in)$$

$$T_B = 0.1253in = T_{Bstress}$$

New Feature C: (Stress)

New Assumptions:

• Due to previous calculated  
(From Fig 1):  $L_c = a + 2b + \text{offset parameter}$

Did NOT account for Length of  
Feature D ( $L_D$ ) Being Implemented  
on top of Feature C:

$$(eq 1) \quad L_{c \text{ new}} = a + 2b + \text{offset parameter} + 2(L_D)$$

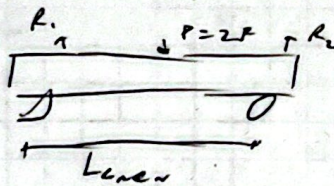
$$L_{c \text{ new}} = (0.498 \text{ in}) + 2(0.9992 \text{ in}) + 0.10 \text{ in} + 2(0.5175 \text{ in})$$

$$\boxed{L_{c \text{ new}} = 3.614 \text{ in}}$$

• Feature C Behaves as Simply Supported  
beam: concentrated load ( $P = 2F$ )  
at midpoint From Feature B  
• Still same  $\text{Inch}^3$  def

From previous Equations Solving For  
Width of C ( $W_{c \text{ stress}}$ ):

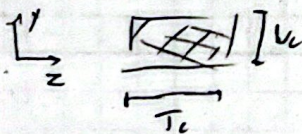
FOD:



Due to Symmetry:

$$R_1 = R_2 = 500 \text{ lb}_f$$

Cross-section:

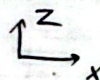


$$\boxed{W_{c \text{ stress}} = \sqrt{\frac{1.5(2F)L_{c \text{ new}} \cdot FOD}{T_c \cdot \sigma_{\text{yield}}}}$$

$$W_{c \text{ stress}} = \sqrt{\frac{1.5(2 \cdot 500 \text{ lb}_f)(3.614 \text{ in})(4)}{(1 \text{ in})(40000 \text{ psi})}}$$

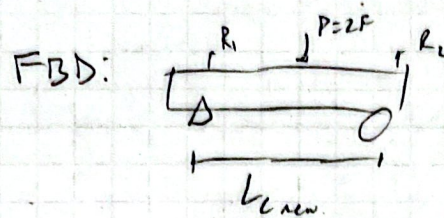
$$\boxed{W_{c \text{ stress}} = 0.7441 \text{ in}}$$

DIAGRAM (IS):  
Feature C



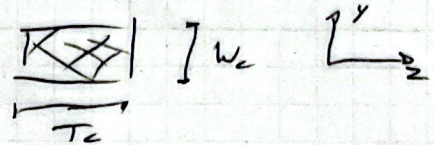
New Feature C (Stiffness);

No New Assumptions Based  
on New Feature C (Stress);



Due to Symmetry:  
 $R_1 = R_2 = 2F$

Cross-section:



From previous Algebraic Solution:

$$W_{c \text{ STIFF}} = \sqrt[3]{\frac{12(2F)(L_{\text{new}})^3}{48(E)(\Delta L_{\text{new}})(T_c)}}$$

$$W_{c \text{ STIFF}} = \sqrt[3]{\frac{12(2 \cdot 5013 \text{ N})(3.6914 \text{ m})^3}{48(10 \cdot 10^6 \text{ psi})(0.005 \text{ m})(1 \text{ m})}}$$

$$W_{c \text{ STIFF}} = 0.2673 \text{ m}$$

Governing  $W_c$ :

$$W_{c \text{ STRESS}} > W_{c \text{ STIFF}}$$

$$(0.7441 \text{ m}) > (0.2673 \text{ m})$$

$$W_c = 0.7441 \text{ m} = W_{c \text{ STRESS}}$$

A6.

## New Feature D (Stress)

New Assumptions:

- Due To Figure 1:  $W_E$

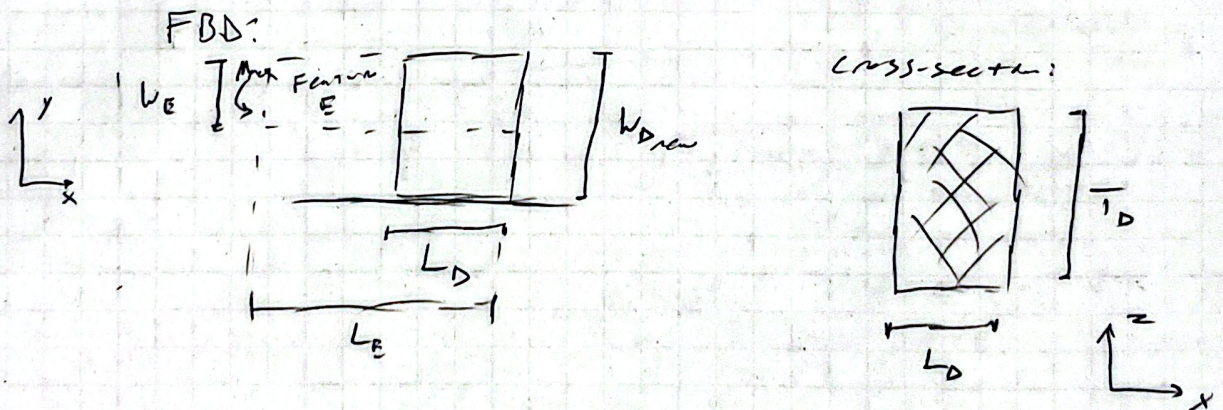
Figure 1 States The rigid beam extends by nominal dimension "C" ( $C = 1.499 \text{ in}$ )

Since  $W_E$  was found before, we can calculate  $W_{D \text{ new}}$ :

$$W_{D \text{ new}} = W_E + C$$

$$W_{D \text{ new}} = 0.5475 \text{ in} + 1.499 \text{ in}$$

$$W_{D \text{ new}} = 2.0465 \text{ in}$$



Using previous Algebraic Expression:  
Solving for  $L_D$ :

$$L_{D \text{ stress}} = \sqrt{\frac{6 \cdot F \cdot L_E \cdot F_{0.5}}{T_D (\sigma_{y \text{ rel}})}}$$

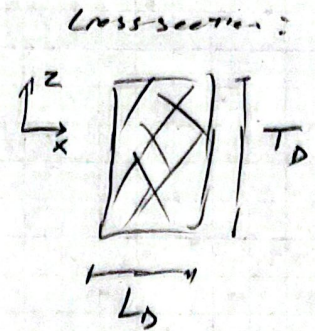
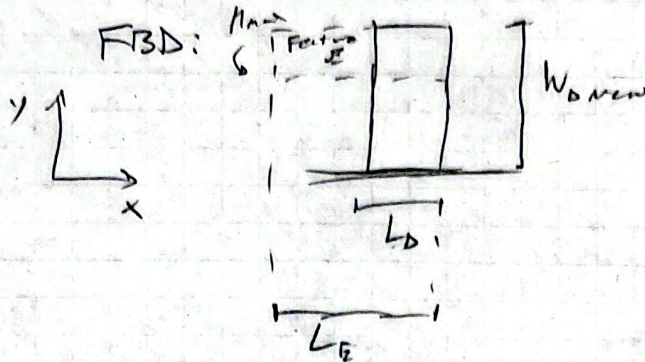
$$L_{D \text{ stress}} = \sqrt{\frac{6 \cdot (500 \text{ lb}) \cdot (0.992 \text{ in}) \cdot (4)}{(1 \text{ in}) (40000 \text{ psi})}}$$

$$L_{D \text{ stress}} = 0.5475 \text{ in}$$

New Feature D (STRA) :

New Assumptions :

No New Assumptions Based  
of a New Feature D (STRA)



Using Previous Algebraic Expression :

$$L_{D \text{ STRA}} = \sqrt[3]{\frac{6(F)(L_E)(W_{D \text{ new}})^2}{E(T_D)(\Delta L_{\text{max}})}}$$

$$L_{D \text{ STRA}} = \sqrt[3]{\frac{6(500 \text{ N})(0.9992 \text{ m})(2.0465 \text{ m})^2}{(10 \cdot 10^6 \text{ psi})(1 \text{ m})(0.005 \text{ m})}}$$

$$L_{D \text{ STRA}} = 0.63087 \text{ m}$$

Governing  $L_D$  :

$$L_{D \text{ STRA}} > L_{D \text{ STRAY}}$$

$$(0.63087 \text{ m}) > (0.5475 \text{ m})$$

$$L_D = 0.63087 \text{ m}$$